

1 Integration of Differential Forms

1.1 Partition of Unity

Definition 1.1 (Support).

Let X be a topological space, and $f : X \rightarrow \mathbb{R}$. Then the support of f is defined as

$$\text{supp } f := \{ x \in X \mid f(x) \neq 0 \}.$$

Theorem 1.1 (Partition of Unity).

Let $\mathcal{M}$ be a C^∞ manifold with dimension m with atlas Φ .

Let $\Phi = \{ \phi_j \mid \phi_j : V_j \rightarrow \phi_j(V_j), j \in J \}$.

Then it is possible to construct a set of C^∞ functions $\rho_j, j \in J$ s.t.

$$1 = \sum_{j \in J} \rho_j, \text{ supp } \rho_j \subseteq V_j$$

1.2 Orientation

1.2.1 Definition

Definition 1.2 (Compatible Coordinate Charts).

Given a manifold $\mathcal{M}$, its two coordinate charts are called compatible (have the same orientation) if,

$$\det J > 0.$$

Where J is the Jacobian matrix ??.

If the manifold has a maximal compatible atlas, then we say the manifold is orientable, and we may call its corresponding orientation positive and denote the atlas Φ_+ .

Theorem 1.2.

A manifold has either no orientation (any atlas is not compatible) or two orientations.

Theorem 1.3 (Orientability and Existence of Forms of Highest Degree).

A manifold is orientable iff there exists a nowhere vanishing differential form of the highest degree.

1.2.2 Positively Oriented Boundary

Definition 1.3 (Positively Oriented Boundary).

Let $\mathcal{M}$ be a orientable C^∞ manifold with dimension m , positively oriented by compatible atlas Φ_+ . Define coordinate charts on $\partial\mathcal{M}$ from Φ as follows,

$$\phi^{\partial\mathcal{M}} : U_\phi \cap \partial\mathcal{M} \rightarrow \mathbb{R}^{m-1},$$

Then $\Phi_+^{\partial\mathcal{M}} := \{ \phi^{\partial\mathcal{M}} \mid \phi \in \Phi_+ \}$ determines an orientation on $\partial\mathcal{M}$, called the positive orientation.

1.3 Pseudoforms (Densities)

1.3.1 Pseudoscalars

Definition 1.4 (Space of Pseudoscalars).

Given a manifold $\mathcal{M}$ and a point p on it. Choose an arbitrary orientation $o_p = \Phi_+$ of $T_p\mathcal{M}$. On the set $\mathbb{R} \times \{o_p, -o_p\}$, define an equivalence relation,

$$\equiv : (c, o_p) \equiv (-c, -o_p).$$

We denote $\tilde{P}_p := \mathbb{R} \times \{o_p, -o_p\} / \equiv$, and call it the space of pseudoscalars. Under a choice of local orientation o , we often denote its element $(c, o) \equiv (-c, -o)$ by $\tilde{c}_o$.

Theorem 1.4 (Pseudoscalar Vector Space).

The space $\tilde{P}_p$ forms a vector space under the operations,

$$\begin{aligned}(a, o) + (b, o) &= (a + b, o), \\ c(a, o) &= (ca, o).\end{aligned}$$

Specifically, $\dim \tilde{P}_p = 1$, since $\forall \tilde{c}_o \in \tilde{P}_p, \exists c \in \mathbb{R}$ s.t. $\tilde{c}_o = c\tilde{1}_o$.

Theorem 1.5 (Pseudoscalar Vector Bundle).

On the space $\tilde{P} = \bigcup_{p \in \mathcal{M}} \tilde{P}_p$, define the canonical projection $\pi : \tilde{P} \rightarrow \mathcal{M}$ by $\tilde{P}_p \mapsto p$.

For any chart $\phi : U \rightarrow \mathbb{R}^m \in \Phi$ of the underlying manifold $\mathcal{M}$, it induces an orientation on $T_p\mathcal{M}$, denoted by o_ϕ . We define the local trivialization by

$$\begin{aligned}\tilde{\phi} : \pi(U) &\rightarrow U \times \mathbb{R} \\ (p, (c, o_\phi)) &\mapsto (p, c).\end{aligned}$$

And this makes pseudoscalars a vector bundle on any manifold $\mathcal{M}$.

Theorem 1.6 (Pseudoscalars Under Coordinate Transformation).

Given two charts $\phi : U \rightarrow \mathbb{R}^m, \psi : V \rightarrow \mathbb{R}^m \in \Phi$ of a manifold $\mathcal{M}$ and a point $p \in U \cap V$, the following relationship holds on the pseudoscalar space $\tilde{P}_p$

Definition 1.4,

$$\tilde{c}_\phi = (\text{sgn det } J)\tilde{c}_\psi,$$

where $\tilde{c}_\phi$ means using the orientation induced by ϕ , and $\text{sgn det } J$ is the sign of the Jacobian matrix ??.

Proof. If $o_\phi = o_\psi$, that is, $\det J > 0$ and their orientations agree, then $(c, o_\phi) \equiv (c, o_\psi)$. But if $o_\phi = -o_\psi$, that is, $\det J < 0$ and their orientations disagree, then $(c, o_\phi) \equiv (-c, o_\psi)$.

This shows that pseudoscalars "flip sign" under charts of different orientation. $\square$

1.3.2 Definition

Definition 1.5 (Pseudoforms).

The space of pseudoforms $\tilde{\Lambda}^k(\mathcal{M})$ is defined by the space of differential forms tensored with (twisted by) the pseudoscalar bundle,

$$\tilde{\Lambda}^k(\mathcal{M}) = \bigwedge^k T^*\mathcal{M} \otimes \tilde{P}(\mathcal{M}).$$

This means, pseudoforms are defined locally as

$$\tilde{\omega} = \tilde{1}_\phi \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}.$$

1.3.3 Pseudoforms and Forms on Orientable Manifolds

Theorem 1.7 (Conversion of Pseudoforms and Forms).

For a pseudoform $\tilde{\omega} \in \tilde{\Lambda}^k(\mathcal{M})$ on an orientable manifold $\mathcal{M}$ of degree m positively oriented by the atlas Φ_+ , the positive orientation is chosen in a continuous manner. Then $\tilde{P}(\mathcal{M})$ is isomorphic to $\mathbb{R}$ by the positive orientation in a continuous manner, so we can say

$$\begin{aligned} \iota : \tilde{\Lambda}^k(\mathcal{M}) &\rightarrow \Lambda^k(\mathcal{M}) \\ \tilde{1}_\phi \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} &\mapsto o(\phi) \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} \end{aligned}$$

defines a smooth differential form on $\mathcal{M}$. Moreover, we can also convert it back,

$$\begin{aligned} \iota^{-1} : \Lambda^k(\mathcal{M}) &\rightarrow \tilde{\Lambda}^k(\mathcal{M}) \\ \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k} &\mapsto \tilde{1}_\phi o(\phi) \omega_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k}. \end{aligned}$$

1.3.4 On Absolute Values

✍ Remark.

Some people would say for $m = \dim \mathcal{M}$, pseudo- m -forms are absolute values of m -forms. This literal definition is doable in our formulation, and is given by the following theorem. □

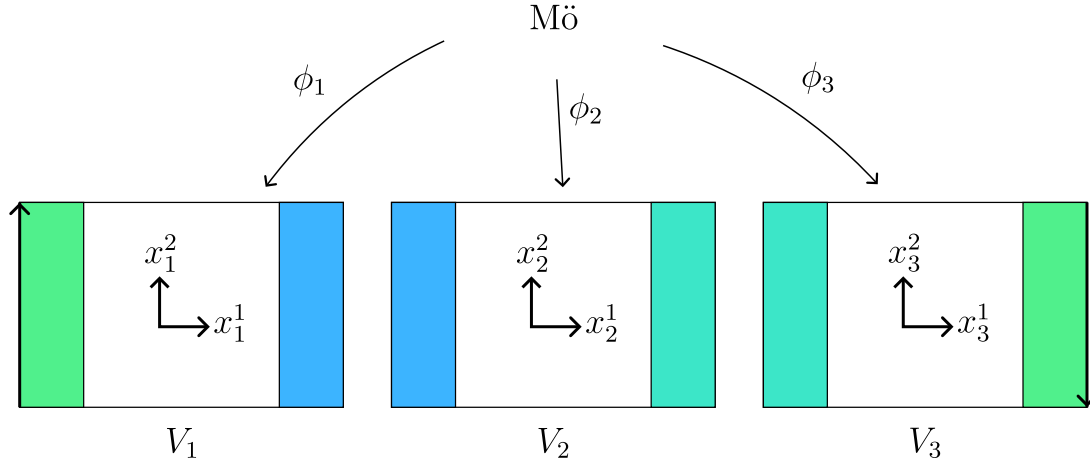
Theorem 1.8.

Given vectors $v_1, \dots, v_m \in T_p \mathcal{M}$, denote their orientation by o_v . Let $\tilde{o}_v(\tilde{\mathbf{I}}_\phi) : \tilde{P}_p \rightarrow \mathbb{R}$ by $(c, o_v) \mapsto c$.

$$\begin{aligned} & (\tilde{o}_v(\tilde{\mathbf{I}}_\phi) f(x) dx^1 \wedge \dots \wedge dx^m)(v_1, \dots, v_m) \\ &= |(f(x) dx^1 \wedge \dots \wedge dx^m)(v_1, \dots, v_m)|. \end{aligned}$$

✈ **Remark.**

To understand these in a deeper level, we should think about why we cannot have a nowhere-vanishing differential form on a non-orientable manifold. Let's try to define a usual form on the Möbius band.



We create a Möbius band by patching three charts together, with overlapping areas indicated by the same color. $V_3 \cap V_1$ are patched together twisted, indicated by the reverse direction of arrows.

Let's say we have the three forms

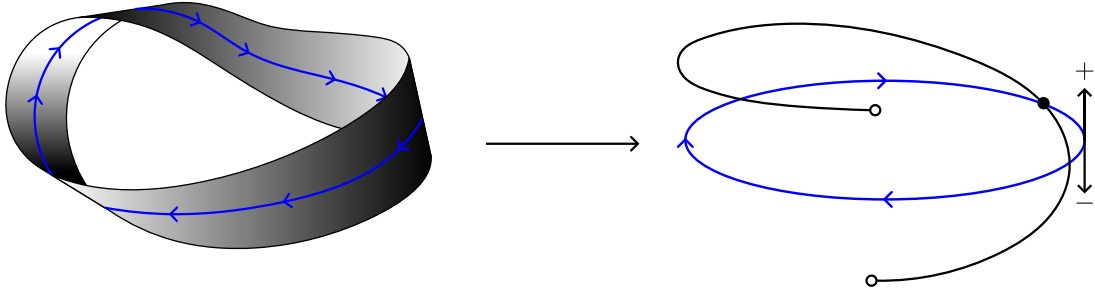
$$\begin{aligned} \omega_1 &= dx_1^1 \wedge dx_1^2 \\ \omega_2 &= f_2 dx_2^1 \wedge dx_2^2 \\ \omega_3 &= f_3 dx_3^1 \wedge dx_3^2 \end{aligned}$$

defined on V_1, V_2, V_3 , respectively, and with ω_1 prescribed to be 1 everywhere.

Then by the pullback criterion, ??, we have

$$\begin{aligned}
\omega_1 &= (\phi_2 \circ \phi_1^{-1})^* \omega_2 \\
&= f_2(\phi_2 \circ \phi_1^{-1}(x_1)) dx_1^1 \wedge dx_1^2 \\
\omega_2 &= (\phi_3 \circ \phi_2^{-1})^* \omega_3 \\
&= f_3(\phi_3 \circ \phi_2^{-1}(x_2)) dx_2^1 \wedge dx_2^2 \\
\omega_3 &= (\phi_1 \circ \phi_3^{-1})^* \omega_1 \\
&= -f_1(\phi_1 \circ \phi_3^{-1}(x_1)) dx_3^1 \wedge dx_3^2.
\end{aligned}$$

But this cannot be, since by evaluating backwards from the last equation, we have $f_3 = -1$, $f_2 = f_3 = -1$, and $f_1 = f_2 = -1$. This contradicts with the given fact that $f_1 = 1$. By this reasoning, we have found an example of the fact that one cannot have a nowhere-vanishing usual form on a non-orientable manifold. Schematically, it looks like this,



where the black line resembles the value of ω . As you can see, it has a mismatch of sign, making it discontinuous. It is possible to resolve this situation by taking the absolute value, but if so, the linearity is broken. Introducing the pseudoscalar allows us to say $\tilde{1}_\phi = -\tilde{1}_\psi$ if their orientations are reversed, so that we can sort of "fold" the ends together, making it continuous again. In terms of coordinates,

$$\begin{aligned}
\tilde{\omega}_3 &= (\phi_1 \circ \phi_3^{-1})^* \tilde{\omega}_1 \\
&= -\tilde{1}_{\phi_1} f_1(\phi_1 \circ \phi_3^{-1}(x_3)) dx_3^1 \wedge dx_3^2 \\
&= \tilde{1}_{\phi_3} f_1(\phi_1 \circ \phi_3^{-1}(x_3)) dx_3^1 \wedge dx_3^2 \\
&= \tilde{1}_{\phi_3} dx_3^1 \wedge dx_3^2.
\end{aligned}$$

□

1.3.5 Metric Volume Form

Definition 1.6 (The Metric Volume Form).

On a smooth Riemannian manifold $\mathcal{M}$ of dimension m with metric g , the metric volume form is the unique pseudo- m -form that gives $\tilde{1}_\phi$ if the vectors are orthonormal and fits the local orientation of ϕ . Especially in 3-dimensional spaces, we denote it as vol .

Theorem 1.9 (Local Representation of Metric Volume Form).

On a smooth Riemannian manifold $\mathcal{M}$ of dimension m with metric g and coordinate chart $\phi = x^1, \dots, x^m$, the metric volume form has local representation

$$\text{vol} = \tilde{1}_\phi \sqrt{\det g} \, dx^1 \wedge \dots \wedge dx^m,$$

where $\det g$ means $\det g_{ij}$, the determinant of the matrix of metric tensor.

Proof. Let $\text{vol} = \tilde{1}_\phi dx^1 \wedge \dots \wedge dx^m$ with the tangent vectors $(\partial_1)_p, \dots, (\partial_m)_p$ orthonormal in the sense of g . Then $g_{\mu\nu} = \delta_{\mu\nu}$ in that basis.

Under another coordinates $\psi = (y^1, \dots, y^m)$, we have

$$\begin{aligned} \text{vol} &= \tilde{1}_\phi \det \left(\frac{\partial x}{\partial y} \right) dy^1 \wedge \dots \wedge dy^m \\ &= \tilde{1}_\psi \left| \det \left(\frac{\partial x}{\partial y} \right) \right| dy^1 \wedge \dots \wedge dy^m. \end{aligned}$$

But

$$\begin{aligned} g &= \delta_{ij} dx^i \otimes dx^j \\ &= \delta_{ij} \frac{\partial x^i}{\partial y^\alpha} \frac{\partial x^j}{\partial y^\beta} dy^\alpha \otimes dy^\beta \end{aligned}$$

by ???. So

$$\begin{aligned} g_{\alpha\beta} &= \sum_k \frac{\partial x^k}{\partial y^\alpha} \frac{\partial x^k}{\partial y^\beta} \\ &= \sum_k (J^{-1})_{k\alpha} (J^{-1})_{k\beta} \\ &= \sum_k ((J^{-1})^\top)_{\alpha k} (J^{-1})_{k\beta} \\ &= ((J^{-1})^\top J^{-1})_{\alpha\beta}. \end{aligned}$$

Thus

$$\det g_{\alpha\beta} = \det((J^{-1})^\top J^{-1}) = (\det(J^{-1}))^2 = \left| \det \frac{\partial x}{\partial y} \right|^2$$

and the proof is completed. $\square$

1.4 Integration of Forms of Highest Degree

Definition 1.7 (Integration).

Let $\mathcal{M}$ be a paracompact C^∞ manifold of dimension m . Choose a C^∞ partition of unity $\rho_j, j \in J$ of $\mathcal{M}$ s.t. $\text{supp } \rho_j \subseteq U_{\phi_j} := U_j$.

Let a pseudo- m -form $\tilde{\omega} \in \tilde{\Lambda}^m(\mathcal{M})$ has local expression $\tilde{\omega}_{\phi_j} = \tilde{1}_{\phi_j} f_j(x) dx_j^1 \wedge \cdots \wedge dx_j^m$.

$$\int_{\mathcal{M}} \tilde{\omega} = \sum_{j \in J} \int_{(-\infty, 0] \times \mathbb{R}^{m-1}} (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m$$

if the finite sum exists and has the same value for all choices of ρ_j and ϕ_j .

✍ Remark.

The following theorem reveals why we integrate pseudoforms, not usual forms.

☞

Theorem 1.10 (Criterion of Existence of Integral).

If $\text{supp } \tilde{\omega}$ is compact, then $\int_{\mathcal{M}} \tilde{\omega}$ exists.

Proof. Let two sets of coordinate charts be

$$\begin{aligned} \phi_j &: U_j \rightarrow V_j, j \in J \\ \phi'_k &: U'_k \rightarrow V'_k, k \in K. \end{aligned}$$

And coresponding partition of unity be ρ_j, ρ'_k .

(The goal) Show

$$\begin{aligned} & \sum_{j \in J} \int (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ &= \sum_{k \in K} \int (\rho'_k \circ \phi'_k{}^{-1})(x') f'_k(x') dx'^1 \dots dx'^m \end{aligned}$$

(Split using ρ'_k)

$$\begin{aligned}\int_{\mathcal{M}} \tilde{\omega} &= \sum_{j \in J} \int (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ &= \sum_{j \in J} \int \sum_{k \in K} (\rho'_k \circ \phi_j^{-1})(x) (\rho_j \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m.\end{aligned}$$

Since the sum is finite, and $\text{supp } \tilde{\omega}$ is compact, and therefore the integral is not improper; thus, there can be no limit or Fubini problems on exchanging sums and integrals. So

$$\int_{\mathcal{M}} \tilde{\omega} = \sum_{j \in J} \sum_{k \in K} \int (\rho_j \rho'_k \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m$$

(Change of variables) First fix j, k .

$$\begin{aligned}& \int (\rho_j \rho'_k \circ \phi_j^{-1})(x) f_j(x) dx^1 \dots dx^m \\ &= \int (\rho_j \rho'_k \circ \phi_j^{-1})(\phi_j \circ \phi'_k{}^{-1}(x')) f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \dots dx'^m \\ &= \int (\rho_j \rho'_k \circ \phi'_k{}^{-1}(x')) f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \dots dx'^m\end{aligned}$$

(Pullback Condition) From ??,

$$\tilde{\omega}_{\phi'_k} = (\phi_j \circ \phi'_k{}^{-1})^* \tilde{\omega}_{\phi_j}.$$

Using **Theorem 1.6**, we see,

$$\begin{aligned}& \tilde{1}_{\phi'_k} f'_k(x') dx'^1 \wedge \dots \wedge dx'^m \\ &= \tilde{1}_{\phi_j} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left(\frac{\partial x^1}{\partial x'^{\ell_1}} dx'^{\ell_1} \right) \wedge \dots \wedge \left(\frac{\partial x^m}{\partial x'^{\ell_m}} dx'^{\ell_m} \right) \\ &= \tilde{1}_{\phi_j} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \sum_{\sigma \in S_m} (\text{sgn } \sigma) \frac{\partial x^1}{\partial x'^{\sigma(1)}} \dots \frac{\partial x^m}{\partial x'^{\sigma(m)}} dx'^1 \wedge \dots \wedge dx'^m \\ &= (\text{sgn } \det J) \tilde{1}_{\phi'_k} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \det \left(\frac{\partial x}{\partial x'} \right) dx'^1 \wedge \dots \wedge dx'^m \\ &= \tilde{1}_{\phi'_k} f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \wedge \dots \wedge dx'^m.\end{aligned}$$

So we see

$$f'_k(x') = f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right|.$$

Therefore, the integral

$$\begin{aligned} & \int (\rho_j \rho'_k \circ \phi'_k{}^{-1}(x')) f_j(\phi_j \circ \phi'_k{}^{-1}(x')) \left| \det \left(\frac{\partial x}{\partial x'} \right) \right| dx'^1 \dots dx'^m \\ &= \int (\rho_j \rho'_k \circ \phi'_k{}^{-1}(x')) f'_k(x') dx'^1 \dots dx'^m \end{aligned}$$

(Closing) By moving the sum wrt $j \in J$ into the integral and using the property of partition of unity, the proof is completed. $\square$

1.5 Integration of Forms of Lower Degree

1.5.1 Pseudoorientation (Transverse Orientation)

$\nearrow$ Remark.

Because pseudoscalars depend on local orientations, we need an extra structure in order to pullback pseudoforms onto a submanifold. $\mathfrak{M}$

Definition 1.8 (Pseudoorientation).

For $\mathcal{Z}$ a submanifold of $\mathcal{M}$, pseudoorientation is a map that for all $p \in \mathcal{Z}$, takes a local orientation of $\mathcal{Z}$ at p (an orientation of $T_p\mathcal{Z}$) to a local orientation of $\mathcal{M}$ at p (an orientation of $T_p\mathcal{M}$) **continuously in p , and takes opposite orientations to opposite orientations.**

$\nearrow$ Remark.

Requiring well-defined surface normal (with a metric, of course) is precisely this criterion for $(m-1)$ dimensional submanifolds, for we can form an orientation of $\mathcal{M}$ from an orientation of $\mathcal{Z}$ (which is essentially $(m-1)$ vectors on $T_p\mathcal{Z}$) and the normal.

This makes Mö integrable as a submanifold of $\text{Mö} \times \mathbb{R}$, since we can choose the surface normal to be positive everywhere. But Mö is not integrable as a submanifold of $\mathbb{R}^3$, since if we try to assign surface normal along the strip, when we get back to where we started, the direction would be reversed. $\mathfrak{M}$

Definition 1.9 (Pullback of Pseudoscalars).

Given a manifold $\mathcal{M}$ and its submanifold $\mathcal{Z}$, and ψ be a chart of $\mathcal{M}$, ϕ a chart of $\mathcal{Z}$. At the point p , if the pseudoorientation i **Definition 1.8** is chosen such that the local orientations $o_\phi \mapsto o_\psi$, then we say $i^* \tilde{1}_\psi = \tilde{1}_\phi$. If $o_\phi \mapsto -o_\psi$, then $i^* \tilde{1}_\psi = -\tilde{1}_\phi$.

1.5.2 Definition**Definition 1.10 (Integration of Lower Degree Forms).**

Let $\mathcal{Z}$ be an pseudooriented C^∞ submanifold of dimension d of $\mathcal{M}$, $i : \mathcal{Z} \rightarrow \mathcal{M}$ be the immersion map.

Let $\omega \in \Lambda^d(\mathcal{M})$, we define

$$\int_{\mathcal{Z}} \omega := \int_{\mathcal{Z}} i^* \omega$$

if it exists.

1.5.3 Stoke's theorem**Theorem 1.11 (Stoke's Theorem).**

If $\mathcal{M}$ is an oriented C^∞ manifold of dimension m and $\omega \in \Lambda^{m-1}(\mathcal{M})$, then

$$\int_{\mathcal{M}} d\omega = \int_{\partial\mathcal{M}} \omega := \int_{\partial\mathcal{M}} i^* \omega,$$

where $i : \partial\mathcal{M} \rightarrow \mathcal{M}$ is just the immersion map, $i : p \mapsto p$.

Proof. (Partition Using Charts) Choose a C^∞ partition of unity **Theorem 1.1** $\rho_j, j \in J$ s.t. $\text{supp } \rho_j$ are compact and $\text{supp } \rho_j \subseteq U_{\phi_j} := U_j$.

Now $\omega = \sum_{j \in J} \rho_j \omega$ is a finite sum. So it suffices to show that, if $\eta \in \Lambda^{m-1}(\mathcal{M})$, $\text{supp } \eta$ compact and $\text{supp } \eta \subseteq U_\phi$ then $\int_{\mathcal{M}} d\eta = \int_{\partial\mathcal{M}} \eta$, and apply $\eta = \rho_j \omega$ for all $j \in J$.

(The integral) Suppose the coordinates of ϕ is labeled $x^1, \dots, x^m$. Locally, let

$$\eta = \sum_{\ell=1}^m f_\ell dx^1 \wedge \dots \wedge dx^\ell \wedge \dots \wedge dx^m,$$

where $f_\ell : (-\infty, 0] \times \mathbb{R}^{m-1} \rightarrow \mathbb{R}$, and it is C^∞ . Then, also locally,

$$\begin{aligned} d\eta &= \sum_{\ell=1}^m \frac{\partial f_\ell}{\partial x^\ell} dx^\ell \wedge dx^1 \wedge \dots \cancel{dx^\ell} \dots \wedge dx^m \\ &= \left(\sum_{\ell=1}^m (-1)^{\ell-1} \frac{\partial f_\ell}{\partial x^\ell} \right) dx^1 \wedge \dots \wedge dx^m. \end{aligned}$$

Then by [Definition 1.7](#),

$$\int_{\mathcal{M}} d\eta = \int_{(-\infty, 0] \times \mathbb{R}^{m-1}} \left(\sum_{\ell=1}^m (-1)^{\ell-1} \frac{\partial f_\ell}{\partial x^\ell} \right) dx^1 \dots dx^m.$$

(To be precise, we need to choose another partition of unity ρ'_j to do integration. But we can just choose it to cover all of $\text{supp } \eta$ and don't care all other parts, so that doesn't matter too much.)

Choose a rectangular region R s.t.

$$\text{supp } \eta \subseteq [a_1, 0] \times \dots \times [a_m, b_m]$$

and define

$$R_\ell := [a_1, 0] \times \dots \cancel{[a_\ell, b_\ell]} \dots \times [a_m, b_m].$$

($\ell = 2, \dots, m$) In this case, by Fubini and FTC,

$$\begin{aligned} &\int_R \frac{\partial f_\ell}{\partial x^\ell} dx^1 \dots dx^m \\ &= \int_{R_\ell} \left(\int_{a_\ell}^{b_\ell} \frac{\partial f_\ell}{\partial x^\ell} dx^\ell \right) dx^1 \dots \cancel{dx^\ell} \dots dx^m \\ &= \int_{R_\ell} \underbrace{(f_\ell(x^1, \dots, b_\ell, \dots, x^m))}_0 - \underbrace{(f_\ell(x^1, \dots, a_\ell, \dots, x^m))}_0 dx^1 \dots \cancel{dx^\ell} \dots dx^m \\ &= 0, \end{aligned}$$

since $\text{supp } \eta \subseteq R$, so on the boundary $f = 0$.

($\ell = 1$) Now the integral has only one term left.

$$\begin{aligned}
\int_{\mathcal{M}} d\eta &= \int_R \frac{\partial f_1}{\partial x^1} dx^1 \dots dx^m \\
&= \int_{R_1} \left(\int_{a_1}^0 \frac{\partial f_1}{\partial x^1} dx^1 \right) dx^2 \dots dx^m \\
&= \int_{R_\ell} (f_1(0, x^2, \dots, x^m) - \underbrace{f_1(a_1, x^2, \dots, x^m)}_0) dx^2 \dots dx^m \\
&= \int_{\mathbb{R}^{m-1}} (f_1 \circ i) dx^2 \dots dx^m \\
&= \int_{\partial \mathcal{M}} i^* \eta.
\end{aligned}$$

■

1.5.4 Integrate Without Partition of Unity

✍ Remark.

The truth is, no one can (or wants) to write down a partition of unity. If one do write it down, integrating it is a big problem.

Fortunately, if one decompose any manifold $\mathcal{M}$ into finitely many submanifolds $\mathcal{M}_i, i \in I$ which intersect only on their boundaries, then $\int_{\mathcal{M}} \tilde{\omega} = \sum_{i \in I} \int_{\mathcal{M}_i} f_i^* \tilde{\omega}$, where f_i is the injective immersion map $f_i : \mathcal{M}_i \rightarrow \mathcal{M}$. □

1.6 Hodge Dual

1.6.1 The Dual Inner Product

Theorem 1.12 (Uniqueness of Dual Inner Product).

Let $\mathcal{M}$ be an m -dimensional Riemannian manifold with metric g . Then for all $k = 1, \dots, m$, g determines a unique inner product $\langle \cdot, \cdot \rangle$ s.t.

$$\langle \omega^1 \wedge \dots \wedge \omega^k, \eta^1 \wedge \dots \wedge \eta^k \rangle = \det \left(\left\langle (\omega^i)^\sharp, (\eta^j)^\sharp \right\rangle_g \right)$$

Proof. (Show dx^I is an orthonormal basis) Let I, J be increasing k -indices, and let $\partial_1, \dots, \partial_k$ be local orthonormal frame. Thus, $g_{ij} = \delta_{ij}$, and $(dx^i)^\sharp = \partial_i$.

If $I = J$,

$$\langle dx^I, dx^I \rangle = \det(\langle \partial_i, \partial_j \rangle) = 1.$$

But if $I \neq J$, $\langle dx^I, dx^J \rangle = 0$ since there would be zero rows.

Now define inner product by

$$\langle \omega, \eta \rangle = \langle \omega_I dx^I, \eta_J dx^J \rangle := \omega_I \eta_J \delta^{IJ}.$$

(I cannot show it satisfy the desired property; but I convinced myself using examples.)

(It remains to show that any other orthonormal frame produces the same inner product. But I hadn't think of a good way.) ▣

Theorem 1.13 (Volume Form Is Of Unit Norm).

On a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , the metric volume form vol is the only m -form that has unit norm with respect to the inner product **Theorem 1.12**.

Proof. Let an m -form $\omega = \lambda dx^1 \wedge \cdots \wedge dx^m$. We wish

$$\begin{aligned} 1 &= \langle \omega, \omega \rangle = \lambda^2 \langle dx^1 \wedge \cdots \wedge dx^m, dx^1 \wedge \cdots \wedge dx^m \rangle \\ &= \lambda^2 \det \left(\left\langle (dx^i)^\sharp, (dx^j)^\sharp \right\rangle \right) \\ &= \lambda^2 \det \left(g_{\mu\nu} (dx^i)^\sharp{}^\mu (dx^j)^\sharp{}^\nu \right). \end{aligned}$$

Now, since $(dx^i)_\mu = \delta^i_\mu$, we know $(dx^i)^\sharp{}^\mu = g^{\mu\nu} \delta^i_\nu = g^{\mu i}$. Moreover, $g_{\mu\nu} g^{\nu\sigma} = \delta_\mu^\sigma$. So

$$\begin{aligned} 1 &= \lambda^2 \det (g_{\mu\nu} g^{\mu i} g^{\nu j}) \\ &= \lambda^2 \det (g^{i\mu} g_{\mu\nu} g^{\nu j}) \\ &= \lambda^2 \det (g^{i\mu} \delta_\mu^j) \\ &= \lambda^2 \det (g^{ij}). \end{aligned}$$

Therefore, $\lambda = \pm \sqrt{\det (g_{ij})}$. ▣

1.6.2 Definition and Uniqueness

Theorem 1.14 (Defining Property of Hodge Star).

On a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , there exists a unique bundle homomorphism $\star : \Lambda^k(\mathcal{M}) \rightarrow \tilde{\Lambda}^{m-k}(\mathcal{M})$ s.t.

$$\omega \wedge \star \eta = \langle \omega, \eta \rangle_g \text{vol}.$$

Proof. (uniqueness) Suppose there exist two such homomorphisms, $\star$ and $\star'$. Then for all $\omega \in \Lambda^k(\mathcal{M})$,

$$\omega \wedge \star \eta = \langle \omega, \eta \rangle_g \text{vol}$$

$$\omega \wedge \star' \eta = \langle \omega, \eta \rangle_g \text{vol}$$

so $\omega \wedge (\star \eta - \star' \eta) = 0$. Thus $\star \eta = \star' \eta$ for all $\eta \in \Lambda^k(\mathcal{M})$.

(existence) We claim for a local orthonormal frame ∂_i ,

$$\star(dx^{i_1} \wedge \dots \wedge dx^{i_k}) = \epsilon^{i_1 \dots i_k}_{j_{k+1} \dots j_m} dx^{j_{k+1}} \wedge \dots \wedge dx^{j_m}.$$

Since

$$\begin{aligned} & (dx^{i_1} \wedge \dots \wedge dx^{i_k}) \wedge \star(dx^{i_1} \wedge \dots \wedge dx^{i_k}) \\ &= \langle dx^I, dx^I \rangle \text{vol} \\ &= \text{vol}, \end{aligned}$$

we need the Levi-Civita symbol to make the indices in correct order. ▣

1.6.3 Coordinate Representation of Hodge Dual

Theorem 1.15 (Hodge Dual of Any Basis).

On a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , the hodge dual satisfies

$$\begin{aligned} & \star(dx^{i_1} \wedge \dots \wedge dx^{i_k}) \\ &= \frac{\sqrt{\det g}}{(n-k)!} g^{i_1 j_1} \dots g^{i_k j_k} \epsilon_{j_1 \dots j_k j_{k+1} \dots j_m} dx^{j_{k+1}} \wedge \dots \wedge dx^{j_m}. \end{aligned}$$

The term $(n-k)!$ is there because we didn't restrict $j_{k+1}, \dots, j_m$ are in order. If we ensure that, then $(n-k)!$ is not there.

Proof. We verify it by calculating

$$(dx^{i_1} \wedge \dots \wedge dx^{i_k}) \wedge \left(\frac{\sqrt{\det g}}{(n-k)!} g^{i_1 j_1} \dots g^{i_k j_k} \epsilon_{j_1 \dots j_k j_{k+1} \dots j_m} dx^{j_{k+1}} \wedge \dots \wedge dx^{j_m} \right).$$

Firstly,

$$\begin{aligned}
\langle dx^{i_1} \wedge \dots \wedge dx^{i_k}, dx^{i_1} \wedge \dots \wedge dx^{i_k} \rangle &= \det \left(\left\langle (dx^i)^\sharp, (dx^j)^\sharp \right\rangle \right) \\
&= \det (g_{\mu\nu} g^{i\mu} g^{j\nu}) \\
&= \det (g^{i\mu} g_{\mu\nu} g^{\nu j}) \\
&= \det (g^{ij}),
\end{aligned}$$

but only $i_1, \dots, i_k$ rows and columns are picked out.

However, $g^{i_1 j_1} \dots g^{i_k j_k} \epsilon_{j_1 \dots j_m}$ are precisely this expression, because $j_{k+1}, \dots, j_m$ cannot contain terms of $i_1, \dots, i_k$, otherwise they wedge to be 0. Therefore, $j_1, \dots, j_k$ are permutations of $i_1, \dots, i_k$, and the Levi-Civita symbol takes care of the $(\text{sgn } \sigma)$. Thus the proof is completed. $\square$

Corollary 1.15.1 (Coordinate Representation of Hodge Dual).

Given a k -form $\omega \in \Lambda^k(\mathcal{M})$ on a Riemannian manifold $\mathcal{M}$ with dimension m and metric g , we have

$$(\star\omega)_{j_{k+1} \dots j_m} = \frac{\sqrt{\det g}}{(n-k)!} g^{i_1 j_1} \dots g^{i_k j_k} \epsilon_{j_1 \dots j_k j_{k+1} \dots j_m} \omega_{i_1 \dots i_k}.$$

1.6.4 Properties of Hodge Dual

Theorem 1.16 (Double Hodge Dual).

For any $\omega \in \Lambda^k(\mathcal{M})$ on a Riemannian manifold $\mathcal{M}$,

$$\star\star\omega = (-1)^{k(m-k)}\omega.$$

✍ Remark.

Note this applies to Euclidean metric only. In other cases, the negative sign may be different. For example, under the Lorentzian metric $g = (-+++)$, the expression is instead

$$\star\star\omega = (-1)^{k(m-k)+1}\omega.$$

$\square$

Proof. It suffices to show for orthonormal frames,

$$\star\star(dx^{i_1} \wedge \dots \wedge dx^{i_k}) = (-1)^{k(n-k)}(dx^{i_1} \wedge \dots \wedge dx^{i_k}).$$

To show this, we verify according to [Theorem 1.14](#),

$$\star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \wedge (-1)^{k(n-k)} (dx^{i_1} \wedge \cdots \wedge dx^{i_k}).$$

We want this to be equal to $\langle \star dx^I, \star dx^I \rangle \text{vol}$. By the exchanging rule ??, we have

$$\begin{aligned} &= (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \wedge \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \\ &= \langle dx^I, dx^I \rangle \text{vol}. \end{aligned}$$

Thus, if $\langle dx^I, dx^I \rangle = \langle \star dx^I, \star dx^I \rangle$, then we are done. I can prove the case when they are orthonormal,

$$\begin{aligned} &\langle \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}), \star (dx^{i_1} \wedge \cdots \wedge dx^{i_k}) \rangle \\ &= \langle \delta^{i_1 j_1} \cdots \delta^{i_k j_k} \epsilon_{j_1 \dots j_n} dx^{j_{k+1}} \wedge \cdots \wedge dx^{j_m}, \delta^{i_1 j'_1} \cdots \delta^{i_k j'_k} \epsilon_{j'_1 \dots j'_n} dx^{j'_{k+1}} \wedge \cdots \wedge dx^{j'_m} \rangle \\ &= 1 \end{aligned}$$

because only the terms which $j_{k+1}, \dots, j_m$ didn't contain $i_1, \dots, i_k$ survive, and Levi-Civita symbol squared is definitely 1. ▣

Theorem 1.17 (Hodge Dual of Constant).

On any Riemannian manifold $\mathcal{M}$,

$$\star 1 = \text{vol}.$$

More generally, for any function $f \in C^\infty(\mathcal{M})$,

$$\star f = f \text{vol}.$$

Theorem 1.18 (Hodge Dual of Flatted Vector Fields).

On any Riemannian manifold $\mathcal{M}$, given a vector field X ,

$$\star X^\flat = X \lrcorner \text{vol}.$$

Proof. Again, we verify [Theorem 1.14](#). In local coordinates,

$$\begin{aligned} \text{vol} &= \sqrt{\det g} dx^1 \wedge \cdots \wedge dx^m \\ X &= X^\mu \partial_\mu \\ X^\flat &= (X^\flat)_\mu dx^\mu = g_{\mu\nu} X^\nu dx^\mu. \end{aligned}$$

Then, using ??,

$$\begin{aligned} & (X^\flat) \wedge (X \lrcorner \text{vol}) \\ &= g_{\mu\nu} X^\nu \sqrt{\det g} \sum_{i=1}^m (-1)^{i-1} X^i dx^\mu \wedge dx^1 \wedge \dots \cancel{dx^i} \dots \wedge dx^m. \end{aligned}$$

But the terms that $\mu \neq i$ vanishes, so

$$\begin{aligned} & (X^\flat) \wedge (X \lrcorner \text{vol}) \\ &= \sqrt{\det g} \sum_{i=1}^m (-1)^{i-1} g_{i\nu} X^\nu X^i dx^1 \wedge \dots \cancel{dx^i} \dots \wedge dx^m \\ &= g_{\mu\nu} X^\mu X^\nu \text{vol} \\ &= \langle X, X \rangle_g \text{vol}. \end{aligned}$$

▣

1.6.5 Other Differential Operators

Definition 1.11 (Codifferential).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , we define the codifferential $\delta : \Lambda^k(\mathcal{M}) \rightarrow \Lambda^{k-1}(\mathcal{M})$.

$$\begin{aligned} \delta &:= (-1)^{k-1} d\star \\ &= (-1)^{m(k+1)+1} \star d\star. \end{aligned}$$

The second expression is given by [Theorem 1.16](#), since the inverse Hodge operator is itself times some negative sign.

Definition 1.12 (Divergence).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , we define the divergence of a vector field to be

$$\begin{aligned} \text{div } X &:= \star d\star X^\flat \\ &= -\delta X^\flat. \end{aligned}$$

The second expression is given by [Definition 1.11](#) and the fact that $k = 1$.

Theorem 1.19.

The divergence of a vector field $\operatorname{div} X$ is the unique scalar that

$$(\operatorname{div} X) \operatorname{vol} = d\star X^\flat.$$

Proof. Note that

$$\operatorname{div} X = -\delta X^\flat = \star^{-1} d\star X^\flat.$$

Therefore,

$$\operatorname{div} X \wedge \star \operatorname{div} X = \langle \operatorname{div} X, \operatorname{div} X \rangle \operatorname{vol}.$$

But since $\operatorname{div} X$ is just a scalar, the above is equivalent to

$$(\operatorname{div} X) \star \operatorname{div} X = (\operatorname{div} X)^2 \operatorname{vol}.$$

If $\operatorname{div} X \neq 0$, then

$$\star \operatorname{div} X = d\star X^\flat = (\operatorname{div} X) \operatorname{vol}.$$

If $\operatorname{div} X = 0$, then by definition $d\star X^\flat = 0$, so the equation trivially holds. $\square$

Theorem 1.20 (Divergence In Local Coordinates).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , the divergence **Definition 1.12** of a vector field X has local expression

$$\operatorname{div} X = \frac{1}{\sqrt{\det g}} \partial_\mu \left(X^\mu \sqrt{\det g} \right).$$

Definition 1.13 (Laplace-de Rham Operator).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , we define the Laplace-de Rham operator to be

$$\Delta := -(d\delta + \delta d),$$

where δ is the codifferential **Definition 1.11**.

✍ Remark.

Both sign conventions are popular. This particular convention makes the local expression in Euclidean space with the standard metric g , reduce to the familiar "Laplacian".

In Minkowski space, the LdR operator is the wave operator $\square$,

the d'Alembertian.

□

Theorem 1.21 (LdR Operator In Local Coordinates).

On a Riemannian manifold $\mathcal{M}$ equipped with metric g , the LdR operator **Definition 1.13** of a function f has local expression

$$\Delta f = \frac{1}{\sqrt{\det g}} \partial_\mu \left(\sqrt{\det g} g^{\mu\nu} \partial_\nu f \right).$$

1.7 Integration of Vector Fields

1.7.1 Line Integrals

1.7.2 Surface Integrals

✍ **Remark.**

In vector calculus, we define the integration of a vector field X over a submanifold S to be

$$\int_{\mathcal{M}} \langle X, N \rangle_g dS,$$

where N is the normal vector of S , $\langle X, N \rangle_g$ is the inner product of the two (induced by metric g), and dS is the (metric) volume form of S . We seek to reformulate this in terms of our language. □

Theorem 1.22 (Hodge Star and Flux).

Let $\mathcal{M}$ be an m -dimensional Riemannian manifold with metric g , and $\mathcal{S}$ be a submanifold of $\mathcal{M}$ of codimension 1. Let X be a vector field on $\mathcal{M}$, dS be the volume forms of $\mathcal{M}$ and $\mathcal{S}$, respectively, and i be the immersion map. Then

$$i^*(\star X^\flat) = \langle X, N \rangle_g dS.$$

Proof. Let $X^\perp = \langle X, N \rangle_g N$, $X^\top = X - X^\perp$. Then

$$X \lrcorner dV = X^\perp \lrcorner dV + X^\top \lrcorner dV.$$

But $X^\perp = \langle X, N \rangle_g N$, so

$$X^\perp \lrcorner dV = \langle X, N \rangle_g N \lrcorner dV = \langle X, N \rangle_g dS.$$

We now show $X^\top \lrcorner dV = 0$. Consider any $X_1, \dots, X_{m-1}$ on $\mathcal{S}$, we see

$$(X^\top \lrcorner dV)(X_1, \dots, X_{m-1}) = dV(X^\top, X_1, \dots, X_{m-1}) = 0$$

since m vectors on $(m-1)$ -dimensional vector spaces are necessarily dependent. $\square$

Theorem 1.23 (Divergence Theorem).

Let $\mathcal{M}$ be an m -dimensional Riemannian manifold with metric g . Denote the metric volume form of $\partial\mathcal{M}$ by dS , then for any compactly supported vector field X , we have

$$\int_{\mathcal{M}} (\operatorname{div} X) \operatorname{vol} = \int_{\partial\mathcal{M}} \langle X, N \rangle_g dS.$$

Proof. By [Theorem 1.19](#), and Stokes theorem [Theorem 1.11](#), we see

$$\int_{\mathcal{M}} (\operatorname{div} X) \operatorname{vol} = \int_{\mathcal{M}} d\star X^\flat = \int_{\partial\mathcal{M}} i^* (\star X^\flat) = \int_{\partial\mathcal{M}} \langle X, N \rangle_g dS.$$

$\square$